

Michael Wang

1031 Walnut Ave 2210, Fremont, CA 94536 | (626) 715-8261 | wangxihao@hotmail.com | <https://oviiy.github.io/>

EDUCATION

University of Southern California (USC)

Los Angeles, CA

Master of Mathematical Finance (GPA 3.8/4)

Spring 2021 – Summer 2022

- Core courses: Corporate Finance, Monte Carlo Simulation, Time Series Analysis, Stochastic Calculus, Risk management, Derivatives pricing, Interest Rate Modeling

University of New South Wales (UNSW)

Sydney, Australia

Bachelor of Electrical Engineering (GPA 3.6/4), Dean's Honor List 2017-2020

Fall 2016 – Spring 2020

- Core courses: Linear algebra, Machine Learning, Optimization, Probability theory, Data Structure
- Exchange Semester at New York University (NYU) 2019

TECHNICAL SKILLS

- **Domain Expertise:** Fixed Income Research, Credit Derivatives Pricing & Modeling, Risk Analytics, Quantitative Model Development & Validation
- **Programming Language:** Python, C/C++, MATLAB
- **AI & Developer Tools:** Claude Code (Skills), Cursor, MCP/API Integration, Deployment Pipelines, CI/CD (Azure DevOps, AWS, Jenkins), Git, Unit/Integration/Regression Testing Design, Visual Studio, LaTeX
- **Data & Analytics:** SQL, Bloomberg (EQS, BDH), Tableau, S3, Power BI, Microsoft Excel (Pivot, VBA, Macros)
- **Statistics Method:** Regression, Clustering, PCA, GARCH-ARIMA, Time Series Analysis
- **Workflows:** Agile/Scrum sprint cycles, Jira/Confluence

WORK EXPERIENCE

Clearwater Analytics

San Jose, CA

Quantitative Analyst (Full-time)

March 2026 – Present

- Research, Develop, validate, and maintain quantitative models and calculations powering Clearwater's Risk & Performance Engineering products, with a focus on fixed income and derivatives analytics.
- Architect and implement production-grade calculation engines for fixed income and derivatives analytics, optimizing for latency and throughput in the Risk & Performance Engineering suite.
- Leverage Claude Code as a primary development tool to accelerate model prototyping, generate test scaffolding, and author validation scripts against expected tie-out outputs for fixed income instruments (e.g., bond pricing, yield curve interpolation, duration/convexity checks, VaR backtesting), reducing sprint QA time by 10+ hours.
- Collaborate cross-functionally with Product and Development teams within agile sprint cycles, participating in sprint planning, standups, and reviews to align priorities and resolve blockers.

MSCI

Berkeley, CA

Quantitative Research Senior Associate (Full-time)

September 2022 – February 2026

- Researched and developed the core C++/Python codebase for 15+ of MSCI's risk analytics asset pricing models, specializing in fixed income and credit derivatives modeling (Bond, Callable, CDS, Credit Index, Credit Index Option, Credit Index Tranche).
- Optimized numerical methods and stochastic calculus implementations (e.g., CDS hazard rate bootstrapping, base correlation curve mapping for standard-index/bespoke tranches, BDT binomial tree for callables) to improve the performance of real-time analytical products.
- Directed cross-functional collaboration with financial engineers, developers, and coverage teams to ensure timely model enhancements driven by client feedback, market convention, and regulatory changes.
- Initiated and developed new research ideas in fixed income and credit derivatives (current run rate of \$80 million), transforming them into production-ready research prototypes incorporated into MSCI's analytical products, such as Bespoke Tranche (Significant Risk Transfer).

- Led research applying the MSCI factor model to fixed-income portfolio construction; evaluated 15+ U.S. equity factor exposures across 10+ years of historical data to identify signals for constructing high-yield corporate-bond baskets.
- Maintained recurring technical interactions with 20+ institutional clients to clarify model assumptions, assess portfolio risk behavior, and explain pricing sensitivities, factor exposures, and stress-testing results.
- Conducted high-quality applied research and wrote research blogs on credit derivatives and fixed-income topics.

Russell Investments

Shanghai, China

Research Analyst Intern

March 2021 – June 2021

- Analyzed and tested the effectiveness of 500+ economy/financial/technical factors for quantitative timing strategy using MACD framework.
- Individually developed and backtested several multi-factor strategies for China's A-share market based on trend following and style rotation, and automated the style-rotation predictive model using Excel (BDH, VBA) and Python. The big/small-cap rotation strategy achieved a profit/loss ratio of 2.53 and 79% accuracy in the testing set (2018-2020).
- Participated in portfolio strategy analysis for China's onshore assets, contributing to the China Team's TAA (Tactical Asset Allocation) framework by integrating the multi-factor strategies developed.

PROJECTS AND PUBLICATIONS

AI-Assisted Fixed Income Instrument Validation, Clearwater Analytics

Spring 2026

Developed a validation framework using Claude Code to automate model testing for fixed income instrument calculations. The workflow uses Claude Code to:

- Generate Python test harnesses from natural-language specifications of bond pricing logic (dirty/clean price, accrued interest, yield-to-maturity, modified duration, convexity).
- Produce tie-out scripts that compare model outputs against Bloomberg reference values and analytically derived benchmarks across instrument types: corporate bonds, Treasuries, MBS, and interest rate swaps.
- Flag calculation regressions by diffing model output snapshots against approved baselines, integrating directly into the Jira QA workflow.

Bespoke Tranche & Base-Correlation Modeling, MSCI

2025-2026

Developed a production-oriented pricing methodology for bespoke synthetic CDO tranches, extending MSCI analytics to customized credit baskets and significant-risk-transfer structures.

- Enhanced and adapted MSCI's existing CDS index-tranche loss-distribution framework for bespoke portfolios, leveraging one-factor Gaussian-copula modeling, conditional default probabilities, probability bucketing, and numerical integration for customized basket composition.
- Researched expected tranche loss and expected tranche notional across attachment and detachment points, supporting protection-leg, premium-leg, present-value, fair-spread, and correlation-calibration analytics.
- Developed a surface-joining framework to map liquid CDX/iTraxx base-correlation quotes to bespoke portfolios, evaluating user-defined curve weighting and expected-tranche-loss mapping for synthetic bespoke correlation curves.

Research Blog: The CDS Market's View on US Default, MSCI

Summer 2023

- Participated in a published blog that discussed the 2023 US government debt ceiling crisis and the implied probability of default of US government from the Credit Default Swap market.
- The blog is focused on discussing the cheapest to deliver bond in the event of CDS auction process. We found a lower market-implied default probability than in 2011 (another debt-ceiling crisis), despite much wider CDS spreads today, highlighting how a simple comparison of CDS spreads may result in misleading inferences.
- The blog is quoted by Reuters and FT: [The CDS Market's View on US Default](#).